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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

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Problem 1: Omitted Variable Bias (OVB)

1(a) Conceptual explanation

The Omitted Variable Bias (OVB) in linear regression analysis develops from the inclusion of a significant explanatory variable that was not included in the model and is correlated with one or more of the remaining regressors (WorldQuant University Module 1). Under these conditions, the Ordinary Least Squares does not provide unbiased and consistent estimates of coefficients.

Thus, the true DGP is

$$Y_t = \beta_0 + \beta_1 X_t + \beta_2 Z_t + \varepsilon_t.$$

If Z_t is omitted, and $\text{Cov}(X_t, Z_t)$ is not equal to zero, the error term picks up the effect of Z_t and thus violates the exogeneity condition $E(\varepsilon_t | X_t) = 0$ (Wooldridge). Again, as repeated in course modules: this kind of violation invalidates estimation and inference since OLS generally attributes part of the effect of Z_t to X_t (WorldQuant University Module 1).

In fact, estimates derived from the poorly specified model would agree with those from the true model just in the event where $\text{Cov}(X_t, Z_t) = 0$ (orthogonality) and also if $\beta_2 = 0$ (the omitted variable effects Y not at all).

1(b) Simulation Design and Intuition.

To illustrate OVB by experiment, we generate data from the following true generation process:

$$Y_t = 1 + 2X_t + 3Z_t + \varepsilon_t$$

where:

- $X_t \sim N(0,1)$
- $Z_t = 0.5X_t + u_t$, ensuring that X_t and Z_t are correlated
- $\varepsilon_t \sim N(0,1)$

Two models are estimated:

1. **Properly specified:** Y_t regressed on X_t and Z_t .
2. **Mis-specified:** Y_t regressed only on X_t .

1(c) Results from Simulation

The table below presents the canonical outputs from the executed notebook

Sample Size	$\hat{\beta}_1$ (Full Model)	$\hat{\beta}_1$ (Omitted Model)
N = 100	1.8979	3.4962
N = 10,000	1.9993	3.4569

1(d) Interpretation and Economic Insight

Empirically, the results will impressively substantiate econometric theory. The value of coefficient X_t in the correctly specified model approaches true value 2 as sample size continues to grow, thus demonstrating the constancy of OLS when its assumptions hold (WorldQuant University Module 1). The somewhat poor condition assessed in the model has a positive bias that is upright and stuck in every case of very large sample sizes. Thus, a very important lesson: omitted variable bias will not vanish asymptotically. Larger datasets reduce variance but do not fix structural misspecifications. In practical, that stresses the importance of theory in model specification; if relevant but unobserved variables exist, then one has to use methods like instrumental variable or fixed effects (Wooldridge).

Problem 2: Sensitivity of OLS to Outliers

2(a) Why OLS is sensitive to outliers

Ordinary Least Squares determines coefficients of regression by reducing to the least sum of square residue. The process of squaring such residuals assigns a proportionately higher weight to an observation with a larger residual during the minimization process (WorldQuant University Module 2). Hence, one such extreme observation is capable of significantly distorting the fitted regression line. Econometrically, this sensitivity arises because OLS is optimal only under the classical assumptions of normally distributed errors within equal variance. OLS is unbiased in theory once such assumptions do not hold in practice, but it loses efficiency and reliability within finite samples. Economically, outliers in financial data resulting from crises, reporting errors, or regime changes may mislead (Greene)

2(b) Simulation Evidence

To illustrate this sensitivity, we estimate a simple linear regression on simulated data and, then, refit the model after adding a single extreme observation. The summary of impacts from the outlier on the regression parameters is presented below

Table 1: Impact of Outlier on OLS Regression Parameter

Metric	No Outlier (Baseline)	With Outlier (Contaminated)	Observed Shift
Intercept ($\hat{\beta}_0$)	-0.5919	-0.3818	Bias introduced
Slope ($\hat{\beta}_1$)	-0.0779	0.9696	Sign Flip (Negative → Positive)
R-squared (R^2)	0.0286	0.4380	Spurious Increase (Masking Effect)

2(c) Interpretation and Economic Meaning

The outlier changes the slope sign from negative to positive and staggers R^2 . Without diagnostics run, an analyst would have concluded that indeed there was a strong positive relation. This emphasizes the importance of influence diagnostics like Cook's Distance and leverage statistics for applied economics and finance (WorldQuant University Module 2). Observations of influence warrant that robust estimation techniques or careful data validation be adopted.

Problem 3: Model Selection

3(a) Conceptual framework

Model selection is primarily considered as a trade-off between goodness of fit and parsimony. The addition of regressors will enhance the in-sample fit almost invariably; however, complex models may be prone to picking up noise rather than real economic relationships. The trade-off is formalized through the use of information criteria such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), which penalize unnecessary complexity (WorldQuant University Module 2). Backward elimination may be considered a compromise between full interpretability and explanatory power in that it starts with the full model and progressively eliminates statistically insignificant variables.

3(b) Empirical application

Backward elimination, validated via the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), was used to compare a reduced model against a full model. Reduced model credibility, measured by AIC and BIC, was found to be lower than full model credibility in the fit-complexity trade-off, confirming that the presence of weakly informative variables reduced the predictive strength of the overall model. Greene advises that such considerations are conducive to interpretability and out-of-sample validity (Greene).

3(c) Interpretation

Backward elimination was employed, and variable X1 was determined to exert either little or no meaningful contribution on the model, on account of a high p-value of 0.8805. The removal of this variable was an improvement judging by both information criteria: AIC from 262.59 to 260.62, with a corresponding reduction in the BIC. The agreement of these criteria suggest that the reduced model provides a better fit by eliminating unnecessary complexity and retaining adequate explanatory power.

3(d) Recommended Course of Action

The presence of unnecessary variables may further cloud the interpretation of a model and diminish its prediction reliability on new data. Thus, we recommend that the selection of predictors be made by formal methods, such as backward elimination, with consideration of multiple information criteria, such as AIC and BIC. This way of working helps avoid overfitting and assures to leave in the model only variables that correctly reflect real economic relationships (WorldQuant University Module 2).

Problem 4: Elasticities

4(a) Elasticity definitions and derivation

Definitions of elasticity and derivations. Elasticity refers to the change in response of Y for any change in X: $E = (dY/dX) \cdot (X/Y)$. Different functional forms yield different elasticity behavior (WorldQuant University Module 3). We here investigate four common specifications:

- 1. Linear:** $Y = \beta_0 + \beta_1 X$
- 2. Log-linear:** $\ln(Y) = \beta_0 + \beta_1 X$
- 3. Log-log:** $\ln(Y) = \beta_0 + \beta_1 \ln(X)$
- 4. Linear-log:** $Y = \beta_0 + \beta_1 \ln(X)$

To mathematically derive the elasticity formulas for all four models, we use Python's SymPy library. The Summary of Functional Form Elasticities is summarized below.

Table 2: Summary of Functional Form Elasticities

Model Type	Example Equation	Derived Elasticity Formula	Elasticity Type
Linear	$Y = 2 + 0.8X$	$0.8 \cdot (X/Y)$	Non-constant (Point Elasticity)
Log-linear	$\ln(Y) = 0.1 + 0.4X$	$0.4 \cdot X$	Non-constant (Depends on X)
Log-log	$\ln(Y) = 0.1 + 0.25\ln(X)$	0.25	Constant

Linear-Log	$Y = 0.15 + 1.2\ln(X)$	$1.2 / (0.15 + 1.2\ln(X))$	Non-constant (Depends on Y)
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4(b) Interpretation

If we now derive the elasticity measure, we find that the log-log model ($\ln(Y) = \beta_0 + \beta_1 \ln(X)$) has a constant elasticity equal to β_1 . Therefore, that percentage changes in X will always cause the same percentage change in Y in that value of X. The other functional forms would not take this effect. In either the linear or semi-log models, elasticity is a function of the value of X or Y, which may not be economically conceivable given certain relationships that should have proportional scaling.

4(c) Recommended Course of Action

For modeling responses that may be considered only proportional, such as demand elasticity or economies of scale, the log-log specification is the most appropriate in such cases. A linear model in these cases indicates that sensitivity varies with scale leading to confusing economic interpretations (WorldQuant University Module 3).

Problem 5: Stationarity and Unit Roots

5(a) Conceptual explanation

A time series is assumed to be weakly stationary if its mean, variance, and autocovariances are constant over time. Many macroeconomic and financial price series violate this condition, following a "random walk," which can lead to spurious regression results if OLS is applied directly (WorldQuant University *Module 4*). We use the **Augmented Dickey-Fuller (ADF)** test to detect the presence of a unit root (non-stationarity).

5(b) Empirical ADF results

The daily adjusted closing price of the S&P 500 index (^GSPC) was subjected to an ADF test. Both the price level and the first difference (daily returns) were tested.

Table 3: Unit Root Test on Equity Data

Series	ADF Test Statistic	p-value	Conclusion
S&P 500 Price Level	0.4489	0.9832	Fail to reject H_0 . Non-stationary (Unit Root).
S&P 500 Returns	-15.8617	0.0000	Reject H_0 . Stationary.

5(c) Interpretation and economic intuition

These results align with financial theory: asset prices follow random walks (non-stationary), while returns are stationary (Hamilton). We are specifically concerned with a **unit root** ($\phi=1$) because it implies shocks have a permanent effect on the level of the series (random walk). We do not typically worry about a root of **1.5** because such a process would be **explosive** (exponentially diverging), which is rarely observed in real economic data. A unit root, however, represents a stochastic trend that is common but problematic for inference. This insight explains why differencing (converting prices to returns) is a standard transformation in financial econometrics.

Problem 6: Structural Breaks

6(a) Methodology

Structural breaks mean that the relationship between the variables evolves over time. If these structural breaks are ignored, results may become biased and subsequent conclusions invalid. This is typically tested using a method known as dummy variable interaction (WorldQuant University Module 5):

$$Y_t = \alpha + \beta t + \delta D_t + \gamma(D_t \cdot t) + \varepsilon_t$$

Where $D_t=1$, after break date and otherwise 0. If γ is statistically significant, then a change in slope is implied. The Chow test, then, is a formal way of comparing restricted (no break) versus unrestricted (break) models.

6(b) Empirical Results

A Chow test was performed to assess whether a structural break at $t=10$ existed. The results presented below compare the restricted (single regime) with unrestricted (two regimes) models

Table 4: Chow Test for Structural Break at $t=10$

Metric	Value	Interpretation
Restricted SSR (No Break)	1617.77	Poor fit (high error)
Unrestricted SSR (Break)	8.69	Excellent fit (low error)
F-Statistic	1480.57	Extremely high test statistic
p-value	0.0000	Reject H_0 (Break Confirmed)

6(c) Economic Relevance and Interpretation

The Cartesian simulation demonstrated how structural breaks could cause significant regime changes to remain neglected. The very high F-statistic (1480.57) and the far-from-zero p-value confirm that the data would follow two distinct trends and not one. These breaks are popularly found in financial markets during times of policy changes, crises, or technological shifts. Modeling breaks makes them more interpretable and improves forecasting, which delineates their importance in applied work (Greene).

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